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PROMISE12 Whole-Gland Benchmark Report

Comparative Evaluation of Four Prostate MRI Segmentation Models

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1. Introduction

This report presents a comparative benchmark of four fixed prostate MRI segmentation models on the PROMISE12 dataset. The goal of this evaluation is to compare the quality of whole-gland prostate segmentation produced by each model after standardizing all predictions into binary masks aligned with the ground-truth image space. The benchmark was carried out in an end-to-end Colab pipeline and includes both quantitative metrics and qualitative visual comparison.

2. Models to be Tested

The following four models were evaluated in this benchmark:

1. **Dzaridis** This model was integrated through the MRI-Prostate-Gland-and-Zone-Segmentor pipeline. For this benchmark, the whole-gland stage was used and the final output was converted into a binary whole-gland mask for evaluation.

2. **DeepInfer** This model was executed through its published prostate segmentation container. The model accepts PROMISE12 MRI volumes and produces a prostate label map that was later standardized to the ground-truth space.

3. **BAMF AIMI** This model is based on an nnU-Net framework trained for prostate segmentation. It produced raw segmentation outputs that were directly used after alignment and binarization.

4. **MONAI prostate_mri_anatomy** This model is a MONAI bundle originally designed for prostate anatomy segmentation. Since the benchmark target is whole-gland segmentation, all non-background labels were merged to form a single binary prostate mask before scoring.

These four models were selected because they represent different deployment styles and segmentation pipelines, which makes the comparison useful both from an accuracy perspective and from an engineering integration perspective.

3. Evaluation Metrics

To compare the performance of the four models, three evaluation metrics were used:

1. **Dice Score** Dice score measures the overlap between the predicted mask and the ground-truth mask. A higher Dice score indicates better agreement between the prediction and the reference segmentation.

2. **Tversky Index** The Tversky index is a generalized overlap metric that allows different weighting of false positives and false negatives. In this benchmark, it was used to penalize false negatives more heavily, making it useful when missing prostate tissue is considered more harmful than slight over-segmentation.

3. **Non-Intersection Percentage** This metric measures the percentage of predicted positive voxels that lie outside the ground-truth region. Lower values are better, since they indicate less over-segmentation outside the true prostate gland.

Together, these metrics provide a balanced view of segmentation quality by measuring overlap, penalizing missed regions, and quantifying unnecessary extra predicted regions.

4. Results

The benchmark report showed that all four models were successfully integrated and evaluated on the PROMISE12 dataset. Out of the 30 total cases, 29 were successfully scored for each model, while one case failed during the standardization stage.

Table 1 summarizes the overall quantitative results.

Table 1: Summary comparison of the four evaluated models

Model	**Scored Cases**	**Failed Cases**	**Dice**	**Tversky**	**NonIntersec**
Dzaridis	29	1	0.9002	0.8844	5.4504
DeepInfer	29	1	0.8376	0.8163	7.3056
BAMF AIMI	29	1	0.9167	0.9172	8.1906
MONAI prostate_mri_anatomy	29	1	0.8154	0.8328	21.9467

Based on the quantitative results, **BAMF AIMI** achieved the best overall segmentation performance in terms of Dice score and Tversky index. **Dzaridis** ranked second and showed strong overall performance with lower non-intersection percentage than BAMF AIMI. **DeepInfer** produced acceptable results but was generally weaker than BAMF AIMI and Dzaridis. **MONAI prostate_mri_anatomy** was the weakest overall model in this benchmark, especially in terms of extra predicted regions outside the ground truth.

Overall, the ranking observed in this benchmark is:

BAMF AIMI > Dzaridis > DeepInfer > MONAI prostate_mri_anatomy

5. Qualitative Visualization

In addition to the numerical metrics, a qualitative visual comparison was included to inspect how well each model captured the prostate shape on a representative PROMISE12 case. The selected case was **Case00**, and the displayed axial slice was **slice 10**.

The visualization is arranged as follows:

- **First row:** Dzaridis
- **Second row:** DeepInfer
- **Third row:** BAMF AIMI
- **Fourth row:** MONAI prostate_mri_anatomy

For each row, the figure shows:

1. the original MRI slice,
2. the ground-truth mask,
3. the model prediction.

This visualization helps confirm the quantitative findings by showing the visual differences in shape agreement, boundary quality, and over-segmentation behavior across the four models.

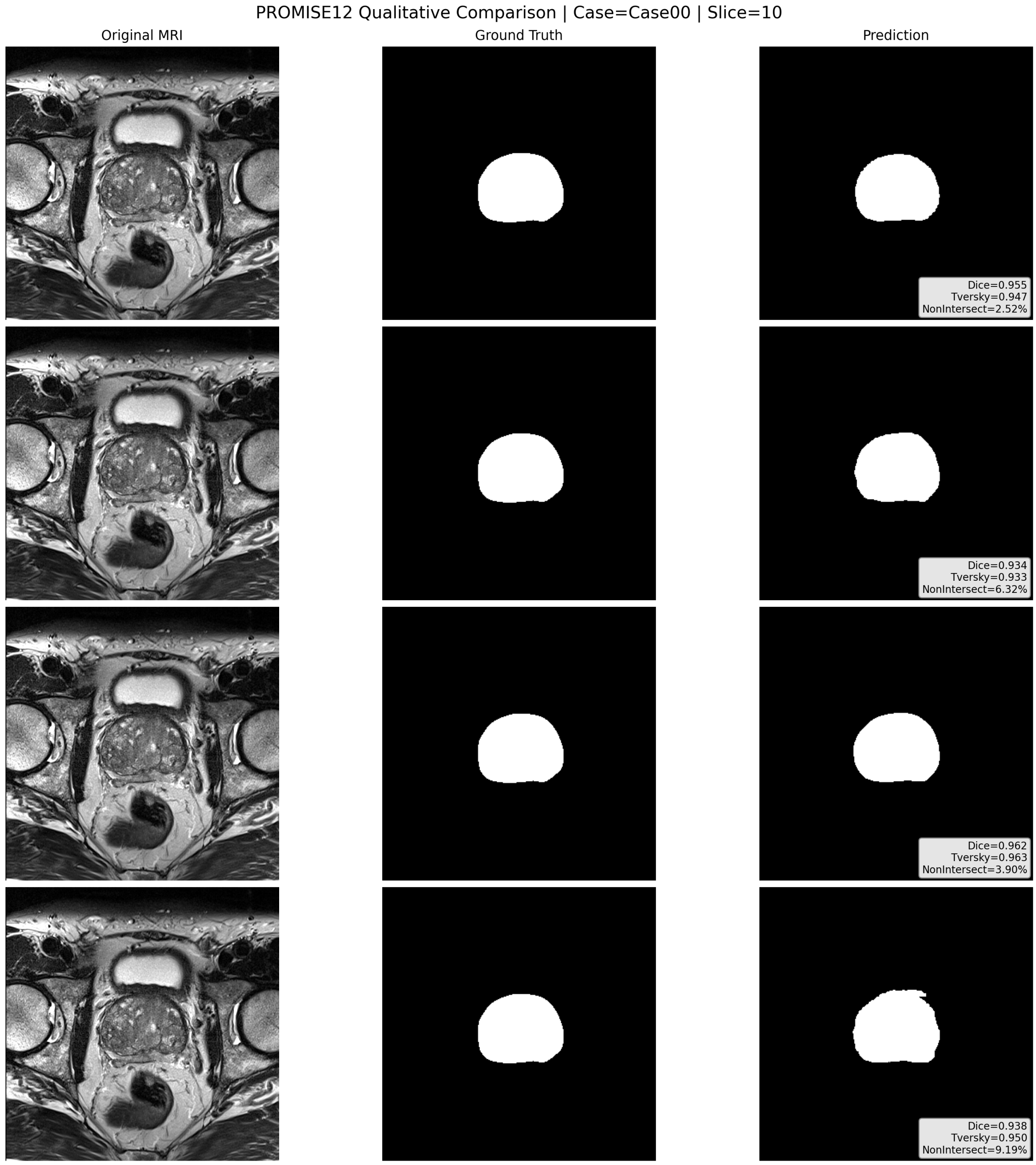

Figure 1: Qualitative comparison on PROMISE12 Case00, slice 10. From top to bottom: Dzaridis, DeepInfer, BAMF AIMI, and MONAI prostate_mri_anatomy.

6. Conclusion

This benchmark compared four prostate MRI segmentation models on the PROMISE12 dataset using unified preprocessing, output standardization, and binary whole-gland evaluation. The results showed that BAMF AIMI achieved the strongest overall performance, followed by Dzaridis, while DeepInfer and MONAI were less accurate overall in this benchmark.